Coherence Change Detection with Sentinel-1 SAR for Conflict Damage Monitoring in the Gaza Strip (2023–2026)

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KEY MESSAGE

Sentinel-1 coherence change detection flags conflict damage across the whole Gaza Strip with **79.5% precision** against UNOSAT — a high-confidence screening tool for humanitarian damage assessment.

1 Why this matters

198,308

damaged structures documented by UNOSAT across the Gaza Strip as of 11 October 2025

- Ground access to conflict zones is restricted
- Very-high-resolution optical imagery is cloud-limited and costly
- Sentinel-1 SAR offers free, all-weather, repeat observations

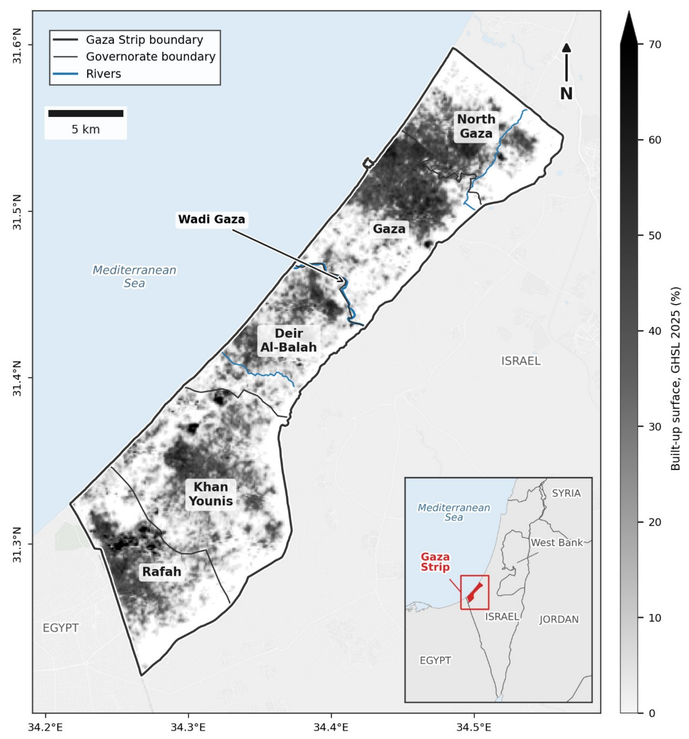
2 The research problem

Low coherence has five causes — only one of them is damage:

1. **Co-registration errors** — sub-pixel misalignment between images
2. **Geometric decorrelation** — different viewing geometry between acquisitions
3. **Temporal decorrelation** — natural change over time (vegetation, soil)
4. **System noise** — sensor thermal and processing artefacts
5. **Real surface change** — damage — the signal we want

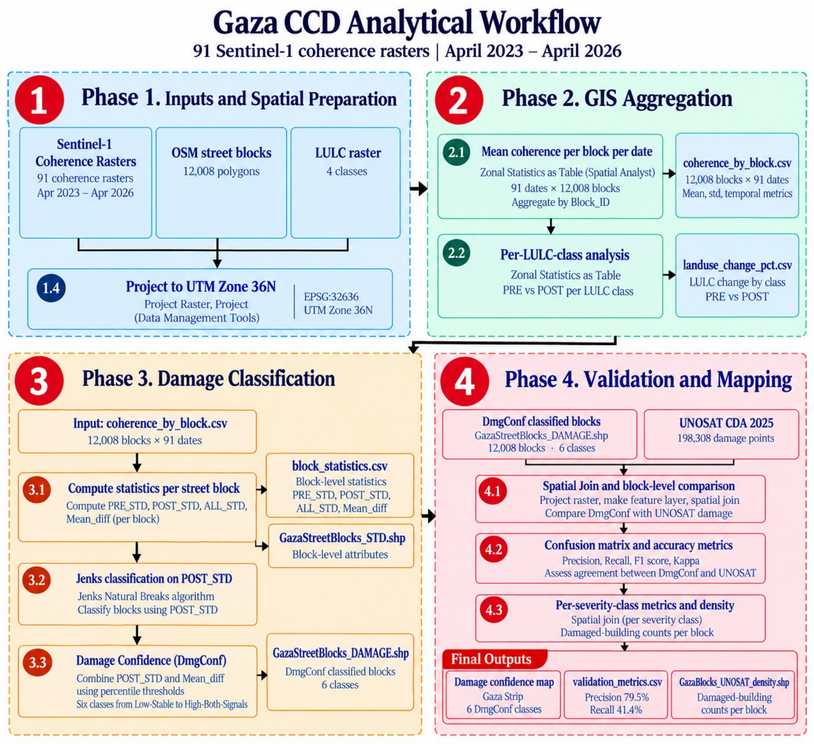
The gap: single-pair CCD cannot separate damage from these sources. This study uses a dense 91-image stack and a combined two-signal metric, validated against UNOSAT.

3 Study area & data

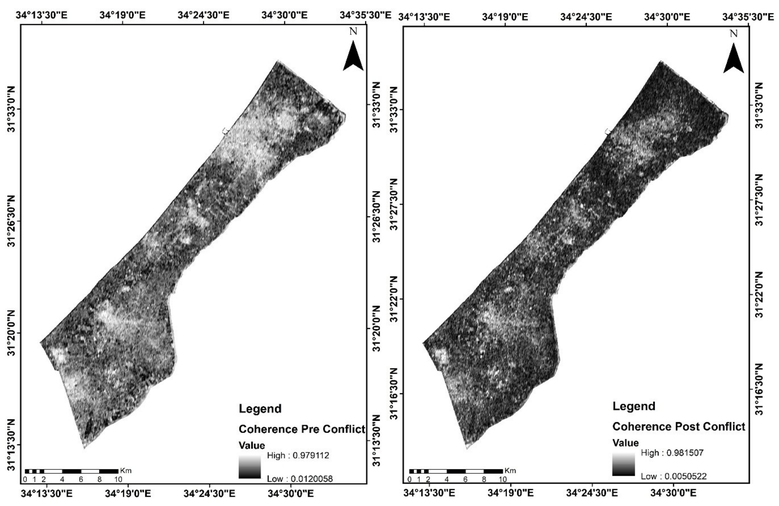


91 Sentinel-1 IW SLC, VV, descending	**3 yrs** April 2023 – April 2026	**16 / 75** pre- / post-conflict pairs (onset 7 Oct 2023)
12,277 OSM-derived street blocks	**198,308** UNOSAT damage points (ground truth)	**365 km²** Gaza Strip, 5 governorates

4 Methodology



5 Coherence decays with time



Same master (28 Feb 2025). Left: 12-day pair, coherence stays high. Right: 408-day pair, coherence collapses widely — mostly natural decorrelation, not damage. A single pair cannot separate damage from time.

6 Damage Confidence metric

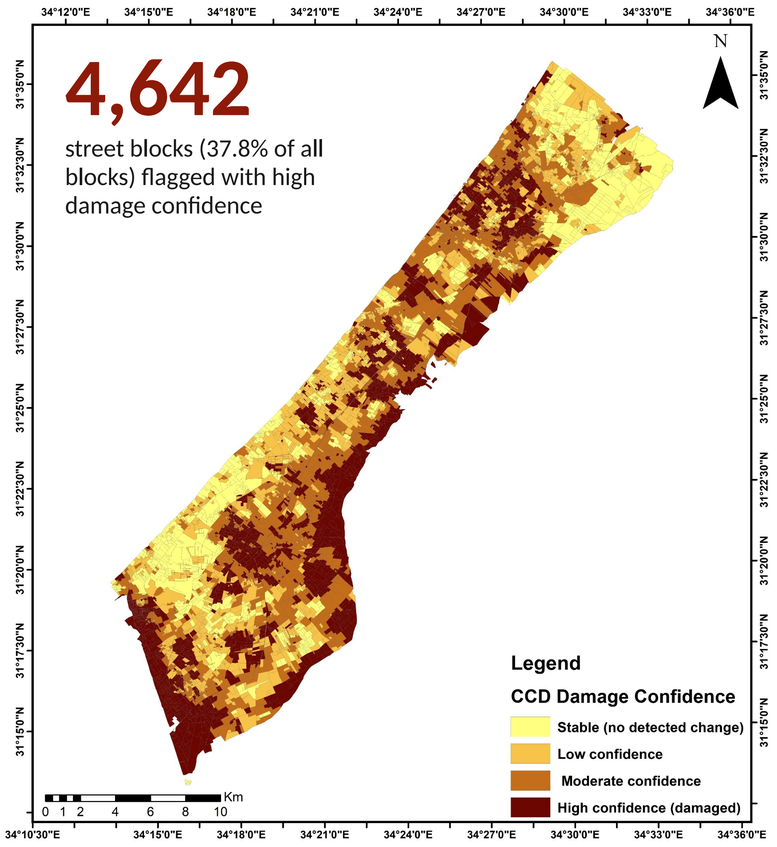
3.3× Median coherence std rises from 0.054 (16 pre-conflict pairs) to 0.176 (75 post-conflict pairs)

Signal 1 · POST_STD	Signal 2 · Mean_diff
Post-event coherence variability → progressive damage that keeps changing	Pre- vs post-event mean shift → one-shot destruction, stable as rubble

Combined into six Damage Confidence classes (p25 / p75 thresholds):

	weight	blocks	share
High — both signals	1.00	**1,361**	**11.1%**
High — progressive	0.85	**1,641**	**13.4%**
High — one-shot	0.85	**1,640**	**13.4%**
Moderate	0.50	3,642	29.7%
Low	0.25	2,377	19.4%
Low — stable	0.10	1,616	13.2%

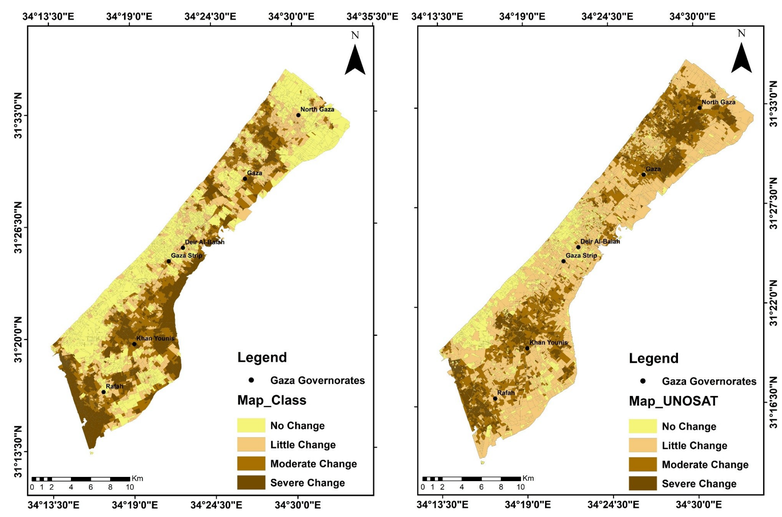
7 Damage Confidence map



Clusters follow documented ground-operation areas:

- **Salah al-Din corridor** · central north–south spine
- **Eastern Gaza City** · Shujaiya, Tuffah, Zeitoun
- **Central camps** · Bureij, Nuseirat, Deir al-Balah
- **Khan Younis** · eastern districts
- **Rafah east** · May 2024 operation area

8 Validation against UNOSAT



Left: CCD damage classes · Right: UNOSAT damage aggregated to the same blocks

n = 12,277 blocks	UNOSAT damaged	UNOSAT not damaged
CCD flagged (High-*)	**3,690**	952
CCD not flagged	5,233	**2,402**

79.5% Precision	**41.4%** Recall	**0.544** F1 score	**0.093** Cohen's κ (slight)

When CCD flags a block, UNOSAT confirms damage in ~4 of 5 cases.

Recall is moderate because 86% of UNOSAT damage was first observed before the Feb 2025 master: early, stabilised rubble shows little variability. κ is low partly because damage prevalence is high.

CONCLUSIONS

01 Territory-scale, 3-year CCD
A 91-image Sentinel-1 stack over the whole Gaza Strip — far denser than the 8–20 images of prior CCD studies.

02 Combined Damage Confidence
POST_STD + Mean_diff capture both progressive and one-shot destruction, which single-metric CCD conflates.

03 Quantitative validation
79.5% precision against 198,308 UNOSAT damage points: a high-precision screening tool, not a complete inventory.

Data & acknowledgements
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